

Maharashtra College of Arts, Science and Commerce

Department of Information Technology and Computer Science

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T.Y.B.Sc (Computer Science) - Project Synopsis (Sem-V)

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Title of the Project : ALTER: AI-Powered Resume Analyzer and Resume Improvement System

Name of the Project Guide : Ms. Saima

Introduction:

ALTER: Land Your Dream Job is an AI-powered web application . The system allows users to securely register and log in, upload one or more PDF resumes, and provide a job description for analysis.

The application is designed to help job seekers improve their resumes according to specific job descriptions. It uses Google's Gemini Large Language Model (LLM) to analyze resumes, identify strengths and weaknesses, highlight missing skills, calculate the ATS (Applicant Tracking System) match percentage, and suggest improvements. It also generates an improved version of the resume tailored to the provided job description.

The application stores users' analysis history in MongoDB, enabling users to review, download, or delete previous analyses. PDF reports of the resume analysis and improved resume can also be generated, making the application useful for students and professionals preparing for internships, placements, and job opportunities.

Objectives:

- To provide an AI-powered platform for analyzing resumes against job descriptions.
- To calculate the ATS compatibility score of a resume.
- To identify missing skills, strengths, and weaknesses in the uploaded resume.
- To generate an improved resume based on the given job description.

- To maintain user authentication and analysis history using MongoDB.
- To allow users to download resume analysis reports and improved resumes in PDF format.

Scope :

- The system allows users to create accounts and securely log in.
- Users can upload one or multiple PDF resumes for analysis.
- The application compares resumes with job descriptions using Google's Gemini AI model.
- The system provides ATS match percentage, resume analysis, and improvement suggestions.
- Analysis history is stored in MongoDB for future reference.
- Users can download resume analysis reports and improved resumes in PDF format.
- The application can be further enhanced by supporting additional resume formats, multiple AI models, and advanced ATS optimization features.

Methodology :

The project follows the Agile Software Development Life Cycle (SDLC).

1. **Requirement Analysis:** Understanding user requirements such as resume upload, AI analysis, ATS matching, authentication, history management, and PDF report generation.
2. **Design:** Designing the Streamlit user interface, MongoDB database structure, authentication module, and integration with the Gemini API.
3. **Development:** Developing the application using Python, implementing resume parsing, AI-based analysis, ATS score calculation, resume improvement, user authentication, and database management.
4. **Testing:** Testing individual modules, verifying AI responses, checking authentication, database operations, and validating PDF generation.

5. **Deployment:** Deploying the Streamlit application with secure API key management and MongoDB connectivity.
6. **Maintenance & Future Enhancement:** Improving AI prompts, supporting DOCX resumes, integrating additional ATS algorithms, and enhancing user experience.

Tools and Technologies

Category	Tools / Technologies Used
Front-End	Streamlit, HTML, CSS
Back-End	Python
Database	MongoDB
AI Model	Google Gemini API (Generative AI)
Development Tools	Visual Studio Code
Libraries	Streamlit, Google Generative AI, PyMuPDF (fitz), Pillow (PIL), PyMongo, ReportLab, python-dotenv, Base64, io, os, time

Resources :

Hardware: Computer/Laptop with Internet Connection

Software:

- Visual Studio Code
- Python
- MongoDB
- Streamlit
- Google Gemini API

Expected Outcome

The **ALTER: AI-Powered Resume Analyzer and Resume Improvement System** provides an intelligent and user-friendly platform for resume evaluation and optimization. Users can upload resumes, compare them with job descriptions, receive AI-generated feedback, calculate ATS match percentages, and generate improved resumes tailored to specific job roles.

The application also offers secure user authentication, maintains resume analysis history, and allows downloading professional PDF reports. The system simplifies the resume optimization process and increases the chances of passing ATS screening, making it a valuable tool for students, fresh graduates, and job seekers.